Grade: Check Plus

The probability distribution of the amount drawn from an ATM at a convenience store is tabulated below. In this table X = the amount withdrawn by a randomly selected customer and f(x) = P[X = x]. A customer can only withdraw the amounts indicated in the first column of the table below.

x	**f(x)**
10	**0.10**
20	**0.15**
40	**0.30**
60	**0.15**
80	**0.25**
100	**0.05**

a. What is the probability that a randomly selected customer withdraws at least $60 from the ATM?

b. Show by calculation that the expected value withdrawn from the ATM is $50 and that the standard deviation of the amount withdrawn from this ATM is $26.27.

c. The manager of this convenience score tells you that an expected value of $50 is not correct because it is impossible to withdraw $50 from the ATM. How would you answer the manager?

d. What is the probability that the amount withdrawn from the ATM will fall within one standard deviation of the mean?

e. What is the probability that the amount withdrawn from the ATM will fall within two standard deviations of the mean?

Problem Set due feb 24th

1. ATM Probability

x	$f(x)$	μ	σ^2
10	.10	1	160
20	.15	3	135
40	.30	12	30
60	.15	9	15
80	.25	20	225
100	.05	5	125
	1.00	50	690

$\mu = x f(x)$

$\sigma^2 = (x-\mu)^2 f x$

$\sigma^2 = x^2 f(x) - \mu^2$

$\sigma^2 = \sqrt{\sigma}$

a.) at least \$60 $x \geq 60$

✓ Qa

.15
.25
.05
.45

45%

b.) $\mu = 50$

✓ Qb

$\sigma = \sqrt{690} = 26.27$

c.) The mean is not a single transaction. This is just the average over time.

✓ Qc

d.) $\mu + \sigma = 50 + 26.27 = 76.27$

✓ Qd

$\mu - \sigma = 50 - 26.27 = 23.73$

$23.73 \leq x \leq 76.27$

$x [40, 60]$

$.3 + .15 = .45 = 45\%$

e.) $\mu + 2\sigma = 102.54$

✓ Qe

$\mu - 2\sigma = -2.54$

$-2.54 \leq x \leq 102.54$ – all values fall w/in range

$x [10, 20, 40, 60, 80, 100]$

$= 100\%$

1. ATM Probability

x	f(x)	Mean	Variance
10	0.10	1	160
20	0.15	3	135
40	0.30	12	30
60	0.15	9	15
80	0.25	20	225
100	0.05	5	125
	1.00	**50**	690
			26.27

a) at least $60 x>=60 0.45 sum of F(x) for 60 and below

b)

Mean =	50
Std. Dev =	26.27

c) The mean is not a single transaction. This just the average over time.

d)

	1 Std. Dev		Valid values between Range:	
Upper	76.27	76	60	0.15
Lower	23.73	24	40	0.30
				0.45

e)

	2 Std. Dev		Valid values between Range:	
Upper	102.54	102	10	0.10
Lower	-2.54	0	20	0.15
			40	0.30
			60	0.15
			80	0.25
			100	0.05
				100%